

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Owned end-to-end delivery of 10+ customer-facing AI features: MCP multi-model pipelines (Claude, GPT-4o, Gemini), Temporal.io workflows, document processing; 7 deployments, 394 commits. - Owned backend: PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry; debugged production issues across full stack.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-based genomic data platform on Google Cloud Run (GCP) with JBrowse 2 genome browser; 100+ researchers, 500GB datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved processing accuracy by 25%. - Automated deployments on AWS and GCP; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

KubeServe	Kubernetes LLM Inference Platform	2026
- Kubernetes LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N pods), Kopf CRD operator for multi-provider LLM serving (Claude, GPT-4o, Gemini); circuit breaker + EWMA LB. - OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 load-tested to 500 VUs; 88 pytest tests.		
AlgoPulse	Agentic AI Platform	2025
- Agentic AI orchestration on Temporal.io and Kafka; sustained 2,000 async inference req/sec across multi-step reasoning pipelines. - Replaced API polling with Supabase Edge Functions streaming AI reasoning loops; no state loss under load.		
VectorLift	Semantic Search Engine	2026
- Semantic search in Python on MS MARCO: BM25, dense bi-encoder (in-batch negatives), cross-encoder re-ranker; Kafka-driven Qdrant corpus indexing pipeline. - Scored NDCG@10, MRR@10, MAP; FastAPI serving with per-stage latency tracking; Streamlit experiment dashboard.		
LLM Eval Harness	Independent Project	2025
- Vendor-neutral benchmark comparing Claude, GPT-4o, Gemini across 2,000 examples and 3 task types in Python. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		

Languages and Technologies

- Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- Backend & APIs:** FastAPI, Node.js, NestJS, PostgreSQL, Redis, REST APIs, Docker, Kafka, Temporal.io
- AI/ML & Agentic:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, LLM evaluation, prompt engineering
- Cloud & DevOps:** AWS, GCP, Kubernetes, KEDA, Helm, Terraform, GitHub Actions, Grafana, OpenTelemetry

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog